



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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SATHYAMANGALAM - 638401 ERODE DISTRICT TAMILNADU INDIA

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22GE004 – BASICS OF ELECTRONICS ENGINEERING

UNIT – I & LIMITATION OF MECHANICAL ENERGY TRANSMISSION

Mechanical energy is the energy possessed by an object due to its motion or position. It is a form of energy that can be transferred between objects through collisions or through the application of external forces.

Mechanical energy is a scalar quantity, meaning it has only magnitude and no direction. The total mechanical energy of an object is the sum of its kinetic energy, which is the energy of motion, and its potential energy, which is the energy of position.

The conservation of mechanical energy states that the total mechanical energy of a closed system remains constant, unless energy is added or removed from the system through work or heat transfer.

Kinetic Energy

Kinetic energy is the energy that an object possesses due to its motion. It is a form of mechanical energy, the energy associated with the movement and position of physical objects. The kinetic energy of an object is equal to the work required to accelerate the object to its current velocity. The formula gives it:

$$\text{Kinetic energy} = \frac{1}{2} * \text{mass} * \text{velocity}^2$$

where mass is the mass of the object and velocity is its speed.

Kinetic energy is a scalar quantity, meaning it has only a magnitude and no direction. It is measured in joules (J) or other units of work. The kinetic energy of an object increases as its mass increases or velocity increases.

For example, a bowling ball moving at high speed has more kinetic energy than a ping-pong ball moving at the same speed. Kinetic energy is important in many real-world situations, such as the motion of a car or the operation of a wind turbine. It is also a key



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concept in the study of thermodynamics since it is related to the temperature and pressure of a gas.

Potential Energy

Potential energy is the energy that an object possesses due to its position within a system. It is a form of mechanical energy, the energy associated with the movement and position of physical objects. Potential energy is stored in an object due to its position within a force field, such as the gravitational field. The formula gives it:

$$\text{Potential energy} = \text{mass} * \text{acceleration due to gravity} * \text{height}$$

where mass is the object's mass, acceleration due to gravity is an object's acceleration due to gravity's force, and height is the object's height above a reference point.

Potential energy is a scalar quantity, meaning it has only a magnitude and no direction. It is measured in joules (J) or other units of work. The potential energy of an object increases as its mass increases or as its height above a reference point increases.

For example, a brick lifted off the ground has more potential energy than a brick already on the ground. Potential energy is important in many real-world situations, such as stretching a rubber band or lifting a heavy object. It is also a key concept in the study of thermodynamics since it is related to the temperature and pressure of a gas.

Conservation of Mechanical Energy

The conservation of mechanical energy is a principle in physics that states that the total mechanical energy of a closed system remains constant as long as no external forces are acting on the system. This means that the sum of an object's kinetic energy (the energy of motion) and potential energy (the energy of position) remains constant as long as no external forces are acting on the object.

For example, consider a ball that is dropped from a height. As the ball falls, it gains kinetic energy due to its motion. At the same time, it loses potential energy due to its position since it is moving closer to the ground. However, the total mechanical energy of the ball remains constant since the decrease in potential energy offsets the increase in



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kinetic energy. Flywheel is another example where the conservation of mechanical energy takes place.

The conservation of mechanical energy is a useful principle in solving problems involving the movement of objects, such as the motion of a ball being thrown or the oscillation of a pendulum. It allows us to predict the behaviour of an object based on its initial conditions, such as its position and velocity, without having to consider all of the forces acting on the object. It's important to note that the conservation of mechanical energy only holds for closed systems, which are systems isolated from external forces. In real-world situations, external forces often act on an object, such as friction or air resistance, can affect the total mechanical energy of the system. These external forces must be considered to accurately predict an object's motion.

Advantages of Conservation of Mechanical Energy

The conservation of mechanical energy is a useful principle in physics with several advantages. Some of the main advantages of the conservation of mechanical energy are:

- **It allows us to predict the behaviour of an object based on its initial conditions:** By understanding the conservation of mechanical energy, we can predict the motion of an object based on its initial position and velocity without having to consider all of the forces acting on the object. This can be especially useful when it is difficult to accurately measure or calculate all the forces acting on an object.
- **It helps us understand the relationships between kinetic and potential energy:** The conservation of mechanical energy helps us understand how kinetic and potential energy are related. For example, we can use the principle of conservation of mechanical energy to understand how an object's kinetic energy changes as it gains or loses potential energy.
- **It simplifies problem-solving:** The conservation of mechanical energy can simplify problem-solving by allowing us to focus on a single quantity (mechanical energy) rather than trying to track multiple quantities, such as kinetic energy, potential energy, and external forces.



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- **It provides a useful tool for understanding real-world phenomena:** The conservation of mechanical energy is a fundamental principle that applies to many real-world phenomena, such as the motion of a ball being thrown or the oscillation of a pendulum. Understanding this principle can help us better understand these phenomena and predict their behaviour.

Limitations of Conservation of Mechanical Energy

The conservation of mechanical energy is a useful principle in physics, but it has some limitations. Some of the main limitations of the conservation of mechanical energy are:

- **It only applies to closed systems:** The conservation of mechanical energy only holds for closed systems, which are systems isolated from external forces. In real-world situations, external forces often act on an object, such as friction or air resistance, can affect the total mechanical energy of the system. These external forces must be considered to accurately predict an object's motion.
- **It does not account for changes in internal energy:** The conservation of mechanical energy only applies to the total mechanical energy of a system, which is the sum of an object's kinetic energy (the energy of motion) and potential energy (the energy of position). It does not account for changes in internal energy, which is the energy associated with the internal structure of an object, such as the energy stored in the bonds between atoms.
- **It does not account for the transfer of energy to or from other forms:** The conservation of mechanical energy only applies to the transfer of mechanical energy within a system. It does not account for the transfer of energy to or from other forms, such as thermal or electrical energy.
- **It is not always accurate in predicting the behaviour of an object:** The conservation of mechanical energy is a useful tool for understanding the behaviour of an object, but it is not always accurate. Other factors that affect an object's motion, such as external forces or internal changes, are not accounted for by the conservation of mechanical energy.

Mechanical Energy Examples



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Mechanical energy is the energy possessed by an object due to its motion or position. Examples of mechanical energy include the kinetic energy of a moving object, the potential energy of an object lifted above the ground, and the elastic potential energy stored in a stretched or compressed spring. The total mechanical energy of an object is the sum of its kinetic and potential energies. Here are some examples of mechanical energy:

- **A roller coaster car** at the top of a hill has potential energy due to its position at a high altitude. As it descends the hill, it converts potential energy into kinetic energy.
- **A baseball thrown through the air** has kinetic energy due to its motion. The faster it is thrown, the more kinetic energy it has.
- **A brick lifted off the ground** and then dropped has potential energy due to its position above the ground. As it falls, it converts potential energy into kinetic energy.
- **A wind turbine** generates mechanical energy as the wind turns its blades, which causes a generator to produce electricity.
- **A swinging pendulum has kinetic energy** due to its motion back and forth. The farther it swings, the more kinetic energy it has.
- **A person lifting a heavy object off the ground** and then setting it down has done work on it, which means that the object has gained mechanical energy.
- **A moving car** has kinetic energy due to its motion. The faster it moves, the more kinetic energy it has.
- **A stretched rubber band** has potential energy due to the tension in the band. When released, it converts potential energy into kinetic energy as it snaps back to its original shape.